Mathematical Reasoning

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Mathematical Reasoning

Statement :

A sentence that is either true or false is said to be a statement, a statement can not be true and false at the same time.

Ex.

(i) "Cuba is the capital of America" a false statement

(ii) "Kathmandu is the capital of Nepal" is a true statement

Any sentence that is a wish or an interrogative or an exclamatory or an imperative, in other words if a sentence is not true or false then that sentence will not be a statement.

(iii) "What have you eaten in the morning?" is an interrogative sentence, hence it is not a statement.

T is truth value of a true statement and F is truth value of a false statement.

Note : It is not possible for a sentence to be a statement to be true and false both at the same time.

Simple statement :

A statement whose truth value (true or false) does not depend on any other statement is called simple statement.

Ex.

(i) "Mumbai is the capital of India" is false statement with truth value F also it's truth value does not depend on any other statement, hence it is a simple statement.

Compound statement :

A statement whose truth value (true or false) depends on any other statement is called compound statement, simply it is the combination of two or more simple statements.

In respect of the compound statement, these simple statements are called sub statements.

For ex.

(i) "If a number is not divisible by 2 then that number is an odd number"

(ii) "Triangle ABC is equilateral if and only if it's three angles are equal"

Illustration :

Q. **Which of the following is NOT a statement ?**

(A) The Sun is a star.

(B) Every quadratic equation has two real roots.

(C) Please get me a glass of water.

(D) $x^2 - 3x + 2 = 0$ implies that $x = -1$ or $x = -2$

Sol: **(C)**

'Please get me a glass of water' is an imperative sentence. So, it is not a statement.

Logical Operator :

Symbols which are used to form compound statement by joining simple statments, are konwn as logical operator.
Some commonly used connectives to connect the simple statements, their symbols and the nature of the compound statement formed after using them is given in the table below.

S.N.	Connectives	Symbol	Use	Operation
(i)	and	$\wedge$	$p \wedge q$	Conjunction
(ii)	or	$\vee$	$p \vee q$	Disjunction
(iii)	not	~ or '	~ p or p'	Negation
(iv)	If.....then.....	$\Rightarrow$ or $\rightarrow$	$p \Rightarrow q$ or $p \rightarrow q$	Implication or Conditional
(v)	If and only if (iff)	$\Leftrightarrow$ or $\leftrightarrow$	$p \Leftrightarrow q$ or $p \leftrightarrow q$	Equivalence or Bi-conditional

Explanation :

(i) $p \wedge q$ means statement p and q

(ii) $p \vee q$ means statement p or statement q

(iii) Not statement p is represented by ~ p

(iv) Statement p then statement q is represented by $p \Rightarrow q$

(v) Statement p if and only if statement q is represented by$p \Leftrightarrow q$

Truth table :

A table which constructed on the basis of truth OR falsety of the simple statement and output of compound statement totally depends on the thruth OR falsety of the simple statements.
Truth table on basis logical operator are given below:

Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Negation

p	$(\sim p)$
T	F
F	T

Conditional

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$ or $p \leftrightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Note: Total number of rows in a compound statement containing n sub statements are 2^n rows.

Illustrations :

Q. **Which of the following statements is not true ?**
(A) Mumbai is the capital of Rajasthan or Maharashtra.
(B) $\sqrt{3}$ is a rational number or an irrational number.
(C) 125 is a multiple of 7 or 8.
(D) A square is a quadrilateral or a regular hexagon

Sol. **(C)**

(a) The component statements of " Mumbai is the capital of Rajasthan or Maharashtra" are
p : Mumbai is the capital of Rajasthan.
q : Mumbai is the capital of Maharashtra.
We note that p is false and q is true, so the compound statement is true.
(b) The component statements of "$\sqrt{3}$ " is a rational or an irrational number" are
p : $\sqrt{3}$ is a rational number.
q : $\sqrt{3}$ is an irrational number.
We note that p is false and q is true, so compound statement is true.
(c) The component statements of "125 is a multiple of 7 or 8" are
p : 125 is a multiple of 7.
q : 125 is a multiple of 8.
We note that p and q both are false statements, so compound statement is false.

(iv) The component statements of "A square is a quadrilateral or a regular hexagon" are
p : A square is a quadrilateral.
q : A square is a regular hexagon.
We note that p is true and q is false. So, the compound statement is true

Q. **Which of the following is wrong statement ?**
(A) The truth table of ~ p ∨ q is same as to p → q
(B) If the truth values of p, q, r are T, F and T respectively, then the truth value of (p ∨ q) ∧ (q ∨ r) is T
(C) ~ (p ∨ q) ≅ ~ p ∧ ~ q
(D) The truth value of p ∧ ~ (p ∨ q) is always T

Sol: **(D)**

For option A we are making the truth table of p→q and ~p∨q:

p	q	~ p	p → q	~p ∨ q
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

We can clearly see the truth value is same for $p \to q$ and $\sim p \vee q$ in all the cases.

Hence, $p \to q$ is logically equivalent to $\sim p \vee q$.

It means option (A) is correct.
Given the condition in B option $p \vee q$ have the truth value as T and $q \vee r$ have the truth value as T also and conjuction of two true statements are true which have T as the truth value.
Hence, option (B) is correct.
We have, $\sim (p \vee q) \cong (\sim p \wedge \sim q)$

So, option, (C) is correct.
$p \wedge \sim (p \vee q)$ will be false when $\sim (p \vee q)$ is false and p is true, for $\sim (p \vee q)$ to be false, $p \vee q$ should be true, and it will be true when p is true and q is false.

Hence, option (D) will be wrong under the above given conditions.

Q. **If p, q and r are given as simple statements, then $(p \wedge q) \wedge (q \wedge r)$ will be true , then**

(A) p, q and r are false **(B) p, q are true and r is false**

(C) p is true and q, r are false **(D) p, q and r are true**

Sol: **(A)**

$(p \wedge q) \wedge (q \wedge r)$ is true means $p \wedge q \cdot q \wedge r$ are both true.

$\Rightarrow p, q, r$ are all true $\therefore$ (A) holds.

Q. **If $p \rightarrow (\sim p \vee \sim q)$ is false, then the truth values of p and q are respectively**

(A) F, F **(B) F, T** **(C) T, T** **(D) T, F**

Sol: **(C)**

p	q	~p	~q	~p ∨ ~q	p→(~p ∨ ~q)
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	T

So, truth values of p and q are T, T.

More Than one Correct Option

Q. **If p and q are simple proposition, then $(\sim p \wedge q) \vee (\sim q \chi p)$ is false when p and q are respectively**

(A) T, T **(B) T, F** **(C) F, F** **(D) F, T**

Sol: **(AC)**

$(\sim p \wedge q) \vee (\sim q \wedge p)$ is false.

$\Rightarrow (\sim p \wedge q)$ is false and $(\sim q \wedge p)$ is also false.

p	q	~p	~q	~p ∧ q	~q ∧ p
T	T	F	F	F	F
T	F	F	T	F	T

F	T	T	F	T	F
F	F	T	T	F	F

Q. **If $(p \wedge \sim q) \wedge (p \wedge r) \rightarrow \sim p \chi q$ is false, then the truth values of p, q and r are respectively**

(A) T, T, T (B) F, F, F (C) T, F, T (D) F, T, F

Sol: **(C)**

Given $(p \wedge \sim q) \wedge (p \wedge r) \rightarrow \sim p \vee q$ is false

$\Rightarrow (p \wedge \sim q \wedge r) \rightarrow \sim p \vee q$ is false

$\Rightarrow \sim(p \wedge \sim q \sim r) \vee (\sim p \vee q)$ is false

$\Rightarrow (\sim p \vee q \vee \sim r) \vee (\sim p \vee q)$ is false

$\Rightarrow \sim p \vee q \vee \sim r$ is false

So, truth values of ~p, q and ~r must be F, F, F.

Thus, truth values of p, q and r must be T, F, T

Assertion and Reason :

Each of these questions contains two statements : statement I (Assertion) and Statement II (Reason) Each of these questions also has four alternative choice, only one of which is the correct answer. You have to select one of the codes (A), (B), (C), (D) given below.

(A) Statement I is true, Statement II is true ; Statement II is a correct explanation for Statement I.

(B) Statement I is true, Statement II is true ; Statement II is not a correct explanation of Statement I.

(C) Statement I true, Statement II is false.

(D) Statement I is false, Statement II is true.

Q. **Let p : Coffee is hot and q : Blood is blue be two statements , then**
Statement I : $p \vee q$: Coffee is hot or blood is blue
Statement II : $p \vee q$: Coffee is not hot or blood is blue

Sol: **(C)**

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p : Ice is cold q:Blood is green
"p ∨ q : Ice is cold or blood is green" is the right statement.

Previous Year's Question

Consider the following three statements: **[JEE MAIN 2019]**
P : 5 is a prime number.
Q : 7 is a factor of 192
R : L.C. M. of 5 and 7 is 35.
Then the truth value of which one of the following statements is true?
(A) (~P) ∨ (Q ∧ R) (B) (P ∧ Q) ∨ (~R)
(C) (~ P) ∧ (~ Q ∧ R) (D) P ∨ (~ Q ∧ R)

Previous Year's Question

Let p, q, r be three statements such that the truth value of (p ∧ q) → (~q ∨ r) is F. Then the truth values of p, q, r are respectively :

[JEE MAIN 2020]

(A) F, T, F (B) T, F, T (C) T, T, F (D) T, T, T

Logical Equivalence :

If two compound statement have same truth value then statements are called as logically equivalent. In truth table, entries in the last column are equal for logically equivalent compound statements. If S_1 and S_2 two statements are logically equivalent then we can write $S_1 = S_2$.

For ex.
Check whether (p → q) and (~p ∨ q) is logically equivalent or not?
To check this we make the truth table for both the compound statements

Concept Building Exercise

Which of the following is/are logically equivalent to (p ∧ q) ?
(A) ~p ∨ ~q (B) p → ~q
(C) ~(~p ∧ ~q) (D) ~ (p → ~ q)

p	q	(~ p)	p → q	~ p ∨ q
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

By the truth table we can clearly see that their truth value is same, hence (p → q) and (~p ∨ q) are logically equivalent

Symbolically, $\boxed{p \to q \equiv \sim p \vee q}$

Illustrations :

Q. **Equivalent statement of the statement "if 8 > 10 then $2^2 = 6$" will be :**

(A) if $2^2 = 6$ then 8 > 10 **(B) 8 < 10 and $2^2 \neq 6$**

(C) 8 < 10 or $2^2 = 6$ **(D) None of these**

Sol. **(D)**

We know that $p \to q \equiv \sim p \vee q$

$\therefore$ equivalent statement will $8 \not> 10$ or $2^2 = 6$

or $8 \leq 10$ or $2^2 = 6$

Q **p : 55 is multiple of 5 and 7.**

q : 81 is a divisible of 3 or 11.

(A) Only p is true **(B) Only q is true**

(C) Both p and q are true **(D) Both p and q are false**

Sol: **(D)**

Statement p is false becuase it is the conjuction of one true (55 is a multiple of 5) and one false (55 is a multiple of 7) statement, hence p is false.

Statement q is false becuase it is the conjuction of one true (81 is a divisible by 3) and one false (81 is divisible by 11) statement, hence q is false.

Hence both statements p and q are false.

Q. **Three statements p, q , r are given as p : Ravi has passed the maths test, q : Ravi is happy, r : It is not true that Ravi has passed the exam therefore he is sad**

Statement I : $r \equiv p \Rightarrow q$

Statement II : The logical equivalent of $p \Rightarrow q$ is $\sim p \vee q$.

Sol: **(D)**

Statement is r : ~p ⇒ q
Hence, Statement I is false for Statement II.
The truth table for p ⇒ q is

p	q	p ⇒ q
T	F	F
T	T	T
F	F	T
F	T	T

Also, the truth table for ~p ∨ q is

p	q	~p	~p ∨ q
T	F	F	F
T	T	F	T
F	F	T	T
F	T	T	T

Hence, we can say that
p ⇒ q ≡ ~p ∨ q

Q. **Check the validity of statements**
Statement-1 : (p ∧ ~ q) ∧ (~ p ∧ q)
Statement-2 : (p → q) ⇔ (~ q → ~ p)

Sol: **(C)**

We know that p → q ⇔ ~ p ∨ q. Therefore,
~ q → ~ p ⇒ ~ (~ q) ∨ ~ p ⇔ q ∨ ~ p ⇔ ~ p ∨ q
Thus, (p → q) ⇔ (~ q → ~ p)
So, statement-2 is true.
Now, (p ∧ ~ q) ∧ (~ p ∧ q)
≡ (~ q ∧ p) ∧ (~ p ∧ q)
≡ ~ q ∧ (p ∧ ~ p) ∧ q
≡ ~ q ∧ c ∧ q
≡ (~ q ∧ q) ∧ c
≡ c ∧ c ≡ c
So, statement-1 is true.

Q. **$p \leftrightarrow q$ is logically equivalent to**

(A) $(p \rightarrow q) \vee (q \rightarrow p)$ **(B)** $(p \wedge q) \rightarrow (q \rightarrow p)$

(C) $(p \wedge q) \rightarrow (q \wedge p)$ **(D) None of these**

Sol: **(D)**

(A)

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \vee (q \rightarrow p)$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	T	T

(B)

p	q	$p \wedge q$	$q \rightarrow p$	$(p \wedge q) \rightarrow (q \rightarrow p)$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	F	F	T	F
F	F	F	T	T	T

(C)

p	q	$p \wedge q$	$q \wedge p$	$(p \wedge q) \rightarrow (q \wedge p)$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	F	T	F
F	F	F	F	T	T

Hence $p \leftrightarrow q$ is not logically equivalent to any of the options.

Q. **The statement $\sim (p \leftrightarrow \sim q)$ is**

(A) equivalent to $\sim p \leftrightarrow q$ **(B) a tautology**

(C) a fallacy **(D) equivalent to $p \leftrightarrow q$**

Sol: **(D)**

See the following truth table.

p	q	~q	p↔~q	~(p↔~q)	(p ↔ q)
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	F	T	T

As the truth table matches, we have the statement ~ (p ↔ ~ q) is equivalent to p ↔ q

TAUTOLOGY AND CONTRADICTION :

(i) Tautology :

If a statement pattern always true, then it is called tautology i.e., its truth value is always T. It is denoted by t.

For ex.

The statement p ∨ ~ (p ∧ q) is a tautology

p	q	p ∧ q	~ (p ∧ q)	p∨ ~ (p ∧ q)
T	T	T	F	T
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

We can see that the truth value of p ∨ ~ (p ∧ q) is T for all p and q, so p ∨ ~(p∧q) is a tautology.

(ii) Contradiction :

If a statement pattern always false, then it is called contradiction. i.e., its truth value always F. It is denoted by c.

For ex.

The statement (p ∨ q) ∧ (~p ∧ ~q) is a contradiction

p	q	~ p	~ q	p ∨ q	(~ p∧ ~ q)	(p ∨ q) ∧ (~ p∧ ~ q)
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	F

It can be observed that truth value of compound statement $(p \vee q) \wedge (\sim p \wedge \sim q)$ is F.

Note :

The negation of a contradiction is a tautology and the negation of a tautology is a contradiction.

Concept Building Exercise



By truth table prove that :

(i) $p \leftrightarrow q \equiv \sim p \leftrightarrow \sim q$

(ii) $p \wedge (\sim p \vee q) \equiv p \wedge q$

(iii) $p \vee (\sim p \vee q)$ is a tautology.

Illustrations :

Q. **The following statement $(p \to q) \to [(\sim p \to q) \to q]$ is**

(A) equivalent to $\sim p \to q$ **(B) equivalent to $p \to \sim q$**

(C) a fallacy **(D) a tautology**

Sol: **(D)**

We have

$(p \to q) \to [(\sim p \to q) \to q]$ simplifying as

$(p \to q) \to ((p \vee q) \to q)$

$(p \to q) \to ((\sim p \wedge \sim q) \vee q)$

$(p \to q) \to ((\sim p \vee q) \wedge (\sim q \vee q))$

$(p \to q) \to (p \to q)$ which is a tautology.

Q. **The statement $(p \to q) \leftrightarrow (\sim p \wedge q)$ is a**

(A) tautology **(B) contradiction**

(C) Neither (A) nor (B) **(D) Both (A) and (B)**

Sol: **(C)**

p	q	~p	p→q	~p ∧ q	(p→q)↔(~p ∧ q)
T	T	F	T	F	F
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	F	F

Q. **If $p \leftrightarrow (\sim p \vee \sim q)$ is true, then the truth values of p and q are respectively**
(A) F, F **(B) F, T** **(C) T, T** **(D) T, F**

Sol: **(D)**

p	q	~p	~q	~p ∨ ~q	p ↔ (~p ∨ ~q)
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	F
F	F	T	T	T	F

So, truth values of p and q are T, F.

Q. **The statement $(p \rightarrow q) \leftrightarrow (\sim p \wedge q)$ is a**
(A) tautology **(B) contradiction**
(C) Neither (a) nor (b) **(D)Both (a) and (b)**

Sol: **(C)**

p	q	$\sim p$	$p \rightarrow q$	$\sim p \wedge q$	$(p \rightarrow q) \leftrightarrow (\sim p \wedge q)$
T	T	F	T	F	F
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	F	F

Q. **The statement $p \rightarrow (q \rightarrow p)$ is equivalent to**
(A) $p \rightarrow (p \leftrightarrow q)$ **(B) $p \rightarrow (p \rightarrow q)$** **(C) $p \rightarrow (p \vee q)$** **(D) $p \rightarrow (p \wedge q)$**

Sol: **(C)**

Let's simplify the statement

$p \rightarrow (q \rightarrow p) \equiv \sim p \vee (q \rightarrow p) \equiv \sim p \vee (\sim q \vee p)$

$\equiv p \rightarrow (p \vee q)$

Previous Year's Question

Which one of the following Boolean expressions is a tautology ? **[JEE MAIN 2019]**

(A) $(p \vee q) \wedge (\sim p \vee \sim q)$ (B) $(p \wedge q) \vee (p \wedge \sim q)$

(C) $(p \vee q) \wedge (p \vee \sim q)$ (D) $(p \vee q) \vee (p \vee \sim q)$

Previous Year's Question

Which one of the following statements is not a tautology ? **[JEE MAIN 2019]**

(A) $(p \wedge q) \rightarrow p$ (B) $(p \wedge q) \rightarrow (\sim p) \vee q$

(C) $p \rightarrow (p \vee q)$ (D) $(p \vee q) \rightarrow (p \vee (\sim q))$

Previous Year's Question

If q is false and $(p \wedge q) \leftrightarrow r$ is true, then which one of the following statements is a tautology? **[JEE MAIN 2019]**

(A) $(p \vee r) \rightarrow (p \wedge r)$ (B) $(p \wedge r) \rightarrow (p \vee r)$ (C) $p \wedge r$ (D) $p \vee r$

Previous Year's Question

Which of the following statements is a tautology ? **[JEE MAIN 2020]**

(A) $\sim(p \wedge \sim q) \rightarrow p \vee q$ (B) $\sim(p \vee \sim q) \rightarrow p \wedge q$

(C) $p \vee (\sim q) \rightarrow p \wedge q$ (D) $\sim(p \vee \sim q) \rightarrow p \vee q$

Previous Year's Question

Which one of the following is a tautology ? **[JEE MAIN 2020]**

(A) $(P \wedge (P \rightarrow Q)) \rightarrow Q$ (B) $P \wedge (P \vee Q)$

(C) $Q \rightarrow (P \wedge (P \rightarrow Q))$ (D) $P \vee (P \wedge Q)$

Previous Year's Question

Which of the following is a tautology? **[JEE MAIN 2020]**

(A) $(p \rightarrow q) \wedge (q \rightarrow p)$ (B) $(\sim p) \wedge (p \vee q) \rightarrow q$

(C) $(q \rightarrow p) \vee \sim(p \rightarrow q)$ (D) $(\sim q) \vee (p \wedge q) \rightarrow q$

Previous Year's Question

Given the following two statements: **[JEE MAIN 2020]**

(S_1) : $(q \vee p) \rightarrow (p \leftrightarrow \sim q)$ is a tautology

(S_2) : $\sim q \wedge (\sim p \leftrightarrow q)$ is a fallacy. Then:

(A) only (S_1) is correct (B) both (S_1) and (S_2) are correct.

(C) only (S_2) is correct (D) both (S_1) and (S_2) are not correct.

Previous Year's Question

The statement $(p \rightarrow (q \rightarrow p)) \rightarrow (p \rightarrow (p \vee q))$ is: **[JEE MAIN 2020]**

(A) equivalent to $(p \vee q) \wedge (\sim p)$ (B) equivalent to $(p \wedge q) \vee (\sim q)$

(C) a contradiction (D) a tautology

ALGEBRA OF STATEMENTS :

If p, q, r are any three statements then the law of algebra of statements are as follow

(i) Idempotent Laws :

(a) $p \wedge p \equiv p$ (b) $p \vee p \equiv p$

i.e., $p \wedge p \equiv p \equiv p \vee p$

p	$(p \wedge p)$	$(p \vee q)$
T	T	T
F	F	F

(ii) Commutative laws :

(a) $p \wedge q \equiv q \wedge p$ (b) $p \vee q \equiv q \vee p$

p	q	$(p \wedge q)$	$(q \wedge p)$	$(p \vee q)$	$(q \vee p)$
T	T	T	T	T	T
T	F	F	F	T	T
F	T	F	F	T	T
F	F	F	F	F	F

(iii) Associative laws :

(a) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ (b) $(p \vee q) \vee r \equiv p \vee (q \vee r)$

p	q	r	$(p \wedge q)$	$(q \wedge r)$	$(p \wedge q) \wedge r$	$p \wedge (q \wedge r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	T	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

Similarly we can prove result (b).

(iv) Distributive laws :

(a) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

(b) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

p	q	r	$(q \vee r)$	$(p \wedge q)$	$(p \wedge r)$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

Similarly we can prove result (b)

(v) De Morgan Laws :

It can be defined as

(a) $\sim (p \wedge q) \equiv \sim p \vee \sim q$ (b) $\sim (p \vee q) \equiv \sim p \wedge \sim q$

p	q	$\sim p$	$\sim q$	$(p \wedge q)$	$\sim (p \wedge q)$	$(\sim p \vee \sim q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

Similarly we can proved result (b)

(vi) Involution laws (or Double negation laws) :

It can be defined as ~(~ p) ≡ p

p	~ p	~ (~ p)
T	F	T
F	T	F

(vii) Identity laws :

Suppose p is a statement, t is tautology and c is contradiction then :

(a) $p \wedge t \equiv p$ (b) $p \vee t \equiv t$

(c) $p \wedge c \equiv c$ (d) $p \vee c \equiv p$

p	t	c	$(p \wedge t)$	$(p \vee t)$	$(p \wedge c)$	$(p \vee c)$
T	T	F	T	T	F	T
F	T	F	F	T	F	F

(viii) Complement Laws :

(a) $p \wedge (\sim p) \equiv c$ (b) $p \vee (\sim p) \equiv t$

(c) $(\sim t) \equiv c$ (d) $(\sim c) \equiv t$

p	~ p	$(p \wedge \sim p)$	$(p \vee \sim p)$
T	F	F	T
F	T	F	T

(ix) Contrapositive laws :

$p \to q \equiv \sim q \to \sim p$

p	q	~ p	~ q	$p \to q$	$\sim q \to \sim p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Concept Building Exercise

Statement $(p \wedge \sim q) \wedge (\sim p \vee q)$ is

(A) a tautology

(B) a contradiction

(C) neither a tautology nor a contradiction

(D) Insufficient data

Illustrations :

Q. **The statement ~ (p ↔ ~ q) is**

(A) equivalent to ~ p ↔ q **(B) a tautology**

(C) a fallacy **(D)equivalent to p ↔ q**

Sol: **(D)**

See the following truth table.

p	q	$\sim q$	$p \leftrightarrow \sim q$	$\sim(p \leftrightarrow \sim q)$	$p \leftrightarrow q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	F	T	T

As the truth table matches, we have the statement

~ (p ↔ ~ q) is equivalent to p ↔ q

Previous Year's Question

If p ⇒ (q ∨ r) is false, then the truth values of p, q, r are respectively : **[JEE MAIN 2019]**

(A) F,T,T (B) T,F,F (C) T,T,F (D) F,F,F

Previous Year's Question

The expression ~(~p → q) is logically equivalent to : **[JEE MAIN 2019]**

(A)~ p ∧ ~q (B) p ∧ ~ q (C) ~ p ∧ q (D) p ∧ q

Previous Year's Question

The Boolean expression ((p∧q)∨(p∨~q))∧(~p∧~q) is equivalent to : **[JEE MAIN 2019]**

(A) p ∧ q (B) p ∧ (~q) (C) (~p) ∧ (~q) (D) p ∨ (~q)

Previous Year's Question

The logical statement **[JEE MAIN 2019]**

[~(~p ∨ q) ∨ (p ∧ r)] ∧ (~q ∧ r)

is equivalent to :

(A) (p ∧ r) ∧ ~ q (B) (~p∧ ~ q) ∧ r (C) ~ p ∨ r (D) (p ∧ ~ q) ∨ r

Previous Year's Question

The Boolean expression~ $(p \Rightarrow (\sim q))$ is equivalent to : **[JEE MAIN 2019]**

(A) $(\sim p) \Rightarrow q$ (B) $p \vee q$ (C) $q \Rightarrow \sim p$ (D) $p \wedge q$

Previous Year's Question

If the truth value of the statement $p \rightarrow (\sim q \vee r)$ is false(F), then the truth values of the statements p, q, r are respectively : **[JEE MAIN 2019]**

(A) F, T, T (B) T, F, ,F (C) T, T, F (D) T, F, T

Previous Year's Question

If the Boolean expression $(p \oplus q) \wedge (\sim p \odot q)$ is equivalent to $p \wedge q$, where $\oplus, \odot \in \{\wedge, \vee\}$, then the ordered pair $(\oplus, \odot)$ is : **[JEE MAIN 2019]**

(A) $(\vee, \wedge)$ (B) $(\vee, \vee)$ (C)$(\wedge, \vee)$ (D) $(\wedge, \vee)$

Previous Year's Question

The logical statement $(p \Rightarrow q) \wedge (q \Rightarrow \sim p)$ is equivalent to : **[JEE MAIN 2020]**

(A) ~ q (B) ~ p (C) p (D) q

Previous Year's Question

If $p \rightarrow (p \wedge \sim q)$ is false, then the truth values of p and q are respectively : **[JEE MAIN 2020]**

(A) F, T (B) T, F (C) F, F (D) T, T

Previous Year's Question

The proposition $p \rightarrow \sim(p \wedge \sim q)$ is equivalent to **[JEE MAIN 2020]**

(A) $(\sim p) \wedge q$ (B) $(\sim p) \wedge q$ (C) q (D) $(\sim p) \vee q$

Negation of compound statements :

If p and q are any two given statements then prove that:

(i) Negation of $(p \wedge q)$ is $\sim p \vee \sim q$:

We will make a truth table for negation of $(p \wedge q)$ and $\sim p \vee \sim q$

p	q	$(p \wedge q)$	$\sim q$	$\sim p$	$\sim (p \wedge q)$	$(\sim p \vee \sim q)$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	F	T	T	T
F	F	F	T	T	T	T

(ii) Negation of $(p \vee q)$ is $\sim p \wedge \sim q$:

We will make a truth table for negation of $(p \vee q)$ and $\sim p \wedge \sim q$

p	q	$(p \vee q)$	$\sim (p \vee q)$	$\sim p$	$\sim q$	$(\sim p \wedge \sim q)$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

(iii) Negation of $p \to q$ is $p \wedge \sim q$:

We will make a truth table for negation of $(p \to q)$ and $p \wedge \sim q$

p	q	$(p \to q)$	$\sim (p \to q)$	$\sim q$	$(p \wedge \sim q)$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	F

(iv) Negation of $p \leftrightarrow q$ is $(p \wedge \sim q) \vee (q \wedge \sim p)$:

Using $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

$\therefore \sim (p \leftrightarrow q) \equiv \sim [(p \to q) \wedge (q \to p)]$

$\equiv \sim (p \to q) \vee \sim (q \to p)$

$\equiv (p \wedge \sim q) \vee (q \wedge \sim p)$

Illustrations :

Q. **The negation of $\sim s \vee (\sim r \wedge s)$ is equivalent to**

(A) $s \vee (r \vee \sim s)$ **(B) $s \wedge r$**

(C) $s \wedge \sim r$ **(D) $s \wedge (r \wedge \sim s)$**

Sol. **(B)**

Using the rules of logic, we have $\sim s \vee (\sim r \wedge s)$

$= (\sim s \vee \sim r) \wedge (\sim s \vee s) = (\sim s \vee \sim r) \wedge t = \sim s \vee \sim r$

Now the negation of above is $\sim (\sim s \vee \sim r) = s \wedge r$

Q. **Let f be a function from a set X to a set Y. Consider the following statements:**
P : For each $x \in X$, there exists unique $y \in Y$ such that $f(x) = y$.
Q : For each $y \in Y$, there exists $x \in X$ such that $f(x) = y$.
R : There exist $x_1, x_2 \in X$ such that $x_1 \neq x_2$ and $f(x_1) = f(x_2)$.
The negation of the statement "f is one-to-one and onto" is
(A) P or not R
(B) R or not P
(C) R or not Q
(D) P and not R

Sol. **(C)**

Negation of 'f is one to one and onto' is R or not Q

Previous Year's Question

For any two statements p and q, the negation of the expression $p \vee (\sim p \wedge q)$ is :

[JEE MAIN 2019]

(A) $p \wedge q$ (B) $p \leftrightarrow q$ (C) $\sim p \vee \sim q$ (D) $\sim p \wedge \sim q$

Previous Year's Question

The negation of the boolean expression $\sim s \vee (\sim r \wedge s)$ is equivalent to :

[JEE MAIN 2019]

(A) r (B) $s \wedge r$ (C) $s \vee r$ (D) $\sim s \wedge \sim r$

Previous Year's Question

Negation of the statement $\sqrt{5}$ is an integer or 5 is irrational' is : **[JEE MAIN 2020]**

(A) $\sqrt{5}$ is irrational or 5 is an integer

(B) $\sqrt{5}$ is not an integer or 5 is not irrational

(C) $\sqrt{5}$ is an integer and 5 is irrational

(D) $\sqrt{5}$ is not an integer and 5 is not irrational

Previous Year's Question

The negation of the Boolean expression $x \leftrightarrow \sim y$ is equivalent to : **[JEE MAIN 2020]**

(A) $(x \wedge y) \wedge (\sim x \vee \sim y)$ (B) $(x \wedge y) \vee (\sim x \wedge \sim y)$

(C) $(x \wedge \sim y) \vee (\sim x \wedge y)$ (D) $(\sim x \wedge y) \vee (\sim x \wedge \sim y)$

Previous Year's Question

The negation of the Boolean expression $p \vee (\sim p \wedge q)$ is equivalent to: **[JEE MAIN 2020]**

(A) $p \wedge \sim q$ (B) $\sim p \vee \sim q$ (C) $\sim p \vee q$ (D) $\sim p \wedge \sim q$

Definition of Converse, Inverse and Contrapositive for of the Conditional Statement ($p \rightarrow q$) :

(i) Converse :

If we interchange the hypothesis and conclusion for a conditional statement then we get the converse of the conditional statement.

Ex: The converse of $p \rightarrow q$ can be defined as $q \rightarrow p$

(ii) Inverse :

To find the inverse of a conditional statement we take the negation of both the hypothesis and conclusion

Ex: The inverse of $p \rightarrow q$ can be defined as $\sim p \rightarrow \sim q$

(iii) Contrapositive :

To find the contrapositiove of a conditional statement we take the negation of both the hypothesis and conclusion and interchange them.

Ex: The contrapositive of $p \rightarrow q$ is defined as $\sim q \rightarrow \sim p$

Illustration :

Q. **Given statement: if $x = 6$ and $y = -12$ then $x - 2y = 30$. The contrapositive of this statement is :**
(A) If $x - 2y \neq 30$ then $x \neq 6$ or $y \neq -12$
(B) If $x - 2y \neq 30$ then $x \neq 6$ and $y \neq -2$
(C) If $x - 2y = 30$ then $x = 6$ and $y = -12$
(D) None of these

Sol: **(A)**

Suppose l, m and n are three statements such that
l : $x = 4$, m : $y = -2$ and n : $x - 2y = 8$
Then given statement in the form l, m, n can be written as $(l \wedge m) \rightarrow n$ and its contrapositive will be $\sim n \rightarrow \sim(l \wedge m)$
i.e., $\sim n \rightarrow (\sim l \vee \sim m)$
i.e., if $x - 2y \neq 30$ then $x \neq 6$ or $y \neq -12$

Q. **The contrapositive of the conditional statement "If Gopal works hard, then he gets a first class" is**
(A) If Gopal gets a first class, then he works hard.
(B) If Gopal works hard, then he gets a first class.
(C) If Gopal does not get a first class, then he does not work hard.
(D) If Gopal does not work hard, then he does not get a first class.

Sol: **(C)**

Let p : Gopal works hard and q : Gopal gets a first class.
Given statement is of the form $p \rightarrow q$
$\therefore$ Its contrapositive is $\sim q \rightarrow \sim p$ i.e., If Gopal does not get a first class, then he does not work hard.

Q. **The converse of the statement "If sun is not shining, then sky is filled with clouds" is**
(A) If sky is filled with clouds, then the sun is not shining.
(B) If sun is shining, then sky is filled with clouds.
(C) If sky is clear, then sun is shining.
(D) If sun is not shining, then sky is not filled with clouds.

Sol: **(A)**

The converse of the given statement is : "If sky is filled with clouds, then the sun is not shining".

Previous Year's Question

The contrapositive of the statement "If you are born in India, then you are a citizen of India", is : **[JEE MAIN 2019]**

(A) If you are born in India, then you are not a citizen of India.
(B) If you are not a citizen of India, then you are not born in India.
(C) If you are a citizen of India, then you are born in India.
(D) If you are not born in India, then you are not a citizen of India.

Previous Year's Question

Contrapositive of the statement
"If two numbers are not equal, then their squares are not equal" is : **[JEE MAIN 2019]**

(A) If the squares of two numbers are not equal, then the numbers are equal
(B) If the squares of two numbers are equal, then the numbers are not equal
(C) If the squares of two numbers are equal, then the numbers are equal
(D) If the squares of two numbers are not equal, then the numbers are not equal

Previous Year's Question

Let A,B,C and D be four non-empty sets. The contrapositive statement of "If $A \subseteq B$ and $B \subseteq D$, then $A \subseteq C$" is : **[JEE MAIN 2020]**

(A) If $A \subseteq C$, then $B \subset A$ or $D \subset B$
(B) If $A \not\subseteq C$, then $A \subseteq B$ and $B \subseteq D$
(C) If $A \not\subseteq C$, then $A \not\subseteq B$ and $B \subseteq D$
(D) If $A \not\subseteq C$, then $A \not\subseteq B$ or $B \not\subseteq D$

Previous Year's Question

The contrapositive of the statement "If I reach the station in time, then I will catch the train" is: **[JEE MAIN 2020]**

(A) If I will catch the train, then I reach the station in time.
(B) If I do not reach the station in time, then I will catch the train.
(C) If I do not reach the station in time, then I will not catch the train.
(D) If I will not catch the train, then I do not reach the station in time.

Previous Year's Question

Contrapositive of the statement : **[JEE MAIN 2020]**
'If a function f is differentiable at a, then it is also continuous at a', is:
(A) If a function f is not continuous at a, then it is not differentiable at a.
(B) If a function f is continuous at a, then it is differentiable at a.
(C) If a function f is continuous at a, then it is not differentiable at a.
(D) If a function f is not continuous at a, then it is differentiable at a.

Previous Year's Question

Consider the statement : "For an integer n, if n^3-1 is even, then n is odd." The contrapositive statement of this statement is: **[JEE MAIN 2020]**

(A) For an integer n, if n is even, then n^3-1 is even
(B) For an integer n, if n is odd, then n^3-1 is even
(C) For an integer n, if n^3-1 is not even, then n is not odd.
(D) For an integer n, if n is even, then n^3-1 is odd